Highlights

Quantifying the Effectiveness of Independent Manual-Valve Verification in Preventing Product-Transfer Risks: A Monte Carlo Process-Safety Assessment

- Independent position sensing cuts modeled undetected misrouting 14-fold
- With uncertain operator response the median reduction decreases to 2.5-fold
- A 10-fold reduction needs about 97% alarm response and bypass below 53%
- Fail-safe blocking neutralizes gateway outages; misidentification erodes benefit
- Pilot power decreases from 95% to 50% when validations cluster by valve